Realization of Perfectly Elastic Collision in the Delay Circuit Model

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Abstract

We extend class B (transverse wave) defined in the previous paper [**?**] to define class E, which holds position information and displacement. After constructing a non-interacting model in which two instances of class E propagate in opposite directions, we define class F, which adds a collision radius r per instance and the perfectly elastic collision formula to the operation function $*$. By treating $w_{\max}$ as an analogue of mass, a perfectly elastic collision satisfying conservation of momentum and energy is realized. The definitions of the delay circuit F_D and the NOT operation circuit are not changed.

1 Introduction

In the previous paper [**?**], oscillation and propagation patterns were constructed using the delay circuit F_D and the NOT operation circuit. In the previous paper [**?**], the framework of superclasses, meta-information, and the operation function $*$ was introduced. In the previous paper [**?**], class B was used to realize a sine wave (transverse wave).

In this paper, we define class E as an extension of class B, and treat the case in which two instances propagate in opposite directions on the same open rope model. We first construct a model without interaction (passing through), and then construct a model that realizes a perfectly elastic collision.

2 Preparation: Summary of Definitions from Previous Papers

This paper presupposes the delay circuit F_D and the NOT operation circuit from the previous paper [**?**], the superclass, meta-information, and operation function $*$ from the previous paper [**?**], and class B (transverse wave) from the previous paper [**?**]. For details of these definitions, refer to the respective previous papers.

3 Definition of Class E

3.1 Information and Meta-information of Class E

We extend class B ([**?**], §3.4) and define class E with position information as follows.

Information of class E:

- i: scalar value (input value; identical to class B)
- w: real value (amplitude; for output; identical to class B)
- x_0: initial position (integer value)
- x: current position (real value)
- dx: displacement (integer value; position change per step)

Meta-information:

- n: counter (number of delays)
- m: number of delays corresponding to one wavelength
- $w_{\max}$: maximum amplitude (positive real value)
- color: attribute for identification

3.2 Operation Function $*$

$$x = n \cdot dx + x_0 \tag{1}$$

$$w = w_{\max} \cdot \sin\left(\frac{n}{m} \cdot 360\text{ř}\right) \tag{2}$$

Each time F_D is passed, n is incremented and the position x and amplitude w are updated.

4 Non-interacting Model (Passing Through)

4.1 Definition of Two Instances

We define two instances of class E.

E_1 **(red):**

E_2 **(blue):**

Information / Meta-information	Value
x_0	0
x	0
dx	+1
m	12
$w_{\max}$	2
color	red

Table 1: Initial conditions of E_1 (red)

Information / Meta-information	Value
x_0	24
x	24
dx	-1
m	12
$w_{\max}$	1
color	blue

Table 2: Initial conditions of E_2 (blue)

4.2 Sequence of Passing Through

E_1 and E_2 execute the operation $*$ independently and do not affect each other. The position at each step is shown below.

At $n = 12$, E_1 and E_2 reach the same position $x = 12$, but since there is no interaction they pass through each other, and at $n = 24$ each reaches the initial position of the other.

4.3 Visualization of Passing Through

The spacetime diagram below shows the trajectories of E_1 (red) and E_2 (blue). The horizontal axis is position x and the vertical axis is the number of delays n (progressing from top to bottom). The size of each point corresponds to the amplitude $|w|$; dark color represents $w > 0$ and light color represents $w < 0$.

5 Model of Perfectly Elastic Collision

5.1 Definition of Class F

We extend class E (§3) to define class F, which has collision functionality. The information and meta-information of class F are identical to those of class E, with the following meta-information added:

- r: collision radius (a real number with $r > 0$; each instance may have a different value)

Collision Detection. For two instances F_1 (collision radius r_1) and F_2 (collision radius r_2), a collision is detected when the following condition is satisfied:

n	x of E_1	x of E_2
0	0	24
1	1	23
2	2	22
$\vdots$	$\vdots$	$\vdots$
12	12	12
$\vdots$	$\vdots$	$\vdots$
24	24	0

Table 3: Positions of E_1 and E_2 at each step (non-interacting model)

$$(x_1 - x_2)^2 < \max(r_1^2,\ r_2^2) \tag{3}$$

Update of dx at Collision. When a collision occurs, dx is updated using the one-dimensional perfectly elastic collision formula, treating $w_{\max}$ as an analogue of mass:

$$dx_1^{\text{new}} = \frac{(w_{\max,1} - w_{\max,2}) \cdot dx_1 + 2\, w_{\max,2} \cdot dx_2}{w_{\max,1} + w_{\max,2}} \tag{4}$$

$$dx_2^{\text{new}} = \frac{(w_{\max,2} - w_{\max,1}) \cdot dx_2 + 2\, w_{\max,1} \cdot dx_1}{w_{\max,1} + w_{\max,2}} \tag{5}$$

This update maintains the following two conserved quantities:

- **Analogue of momentum**: $Q = w_{\max,1} \cdot dx_1 + w_{\max,2} \cdot dx_2$
- **Analogue of energy**: $K = w_{\max,1} \cdot dx_1^2 + w_{\max,2} \cdot dx_2^2$

5.2 Sequence of Collision

Using the same initial conditions as §4.1, and setting $r_1 = r_2 = 1$ for simplicity.

Before collision ($n < 12$): The trajectories are identical to those of E_1 and E_2 (see §4.2).

Collision at $n = 12$: From $(x_1 - x_2)^2 < \max(r_1^2, r_2^2)$, a collision is detected. dx is updated:

$$dx_1^{\text{new}} = \frac{(2-1)(1) + 2(1)(-1)}{2+1} = \frac{-1}{3} = -\frac{1}{3} \tag{6}$$

$$dx_2^{\text{new}} = \frac{(1-2)(-1) + 2(2)(1)}{2+1} = \frac{5}{3} \tag{7}$$

Verification:

$$Q^{\text{new}} = 2 \cdot (-1/3) + 1 \cdot (5/3) = -2/3 + 5/3 = 1 = Q \quad \checkmark \tag{8}$$

$$K^{\text{new}} = 2 \cdot (1/9) + 1 \cdot (25/9) = 27/9 = 3 = K \quad \checkmark \tag{9}$$

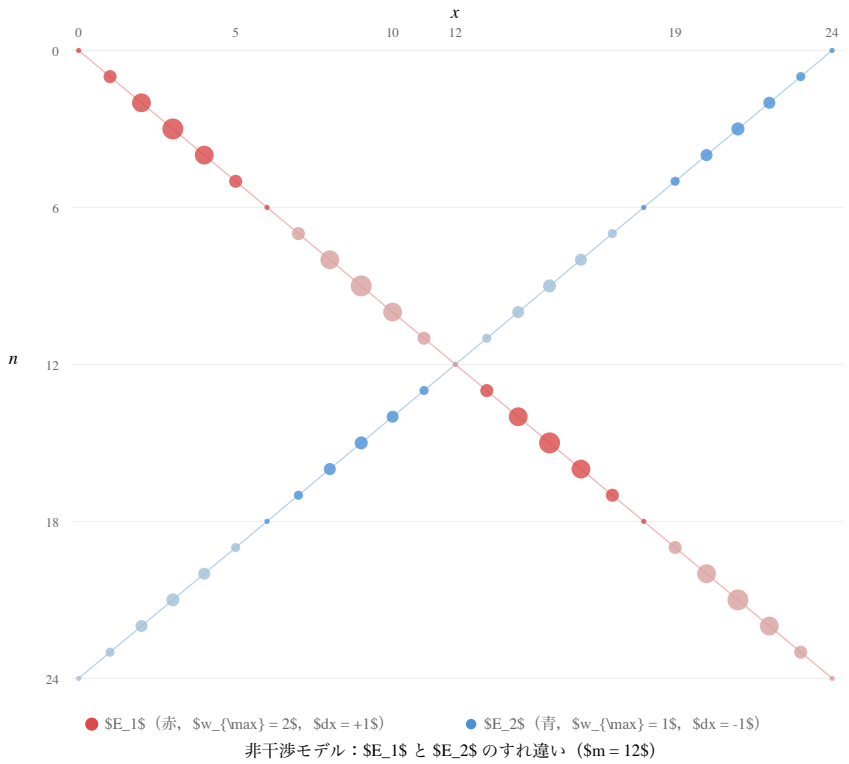

Figure 1: Non-interacting model — passing through of E_1 and E_2 ($m = 12$). At $n = 12$, both pass through the same position $x = 12$, but since there is no interaction they continue straight ahead without any effect on each other.

n	x of F_1	x of F_2
12	12.00	12.00
13	11.67	13.67
14	11.33	15.33
15	11.00	17.00
18	10.00	22.00
21	9.00	27.00
24	8.00	32.00

Table 4: Positions of F_1 and F_2 after the collision

After collision ($n > 12$): The heavier F_1 with $w_{\max} = 2$ slowly returns in the reverse direction ($dx = -1/3$), while the lighter F_2 with $w_{\max} = 1$ is knocked away at high velocity ($dx = 5/3$).

5.3 Visualization of Collision

The spacetime diagram below shows the trajectories of F_1 (red) and F_2 (blue). At the collision point at $n = 12$ (dashed circle), both trajectories bend. The heavier F_1 slowly returns to the left after the collision, while the lighter F_2 is knocked away to the right at high velocity.

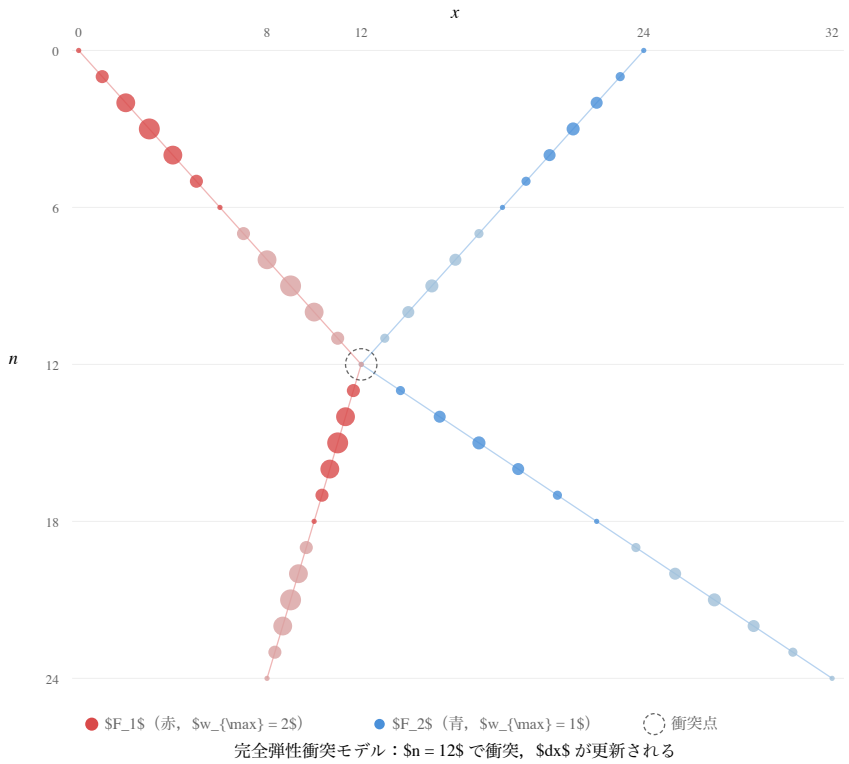

Figure 2: Perfectly elastic collision model — collision at $n = 12$, dx is updated. Comparing with Figure **??** (non-interacting model), the trajectories before the collision ($n < 12$) are identical, but after the collision ($n > 12$) the slopes of the trajectories change to reflect the mass ratio (the ratio of $w_{\max}$).

6 Conclusion

We extended class E (§3) to define instances with position information and displacement. For the case in which two instances propagate in opposite directions on the same space, we first constructed a non-interacting model (§4, passing through), and then realized a perfectly elastic collision by adding a collision radius r and the perfectly elastic collision formula to the operation function $*$ via class F (§5).

In either case, the definitions of the delay circuit F_D and the NOT operation circuit are not changed. The presence or absence of interaction is expressed solely as a difference in the definition of the operation function $*$.

References

[1] Noriaki Kihara, “Information-Theoretic Organization of Information Transmission,” Research Note, April 2026.

[2] Noriaki Kihara, “Organization of Symmetries Inherent in the Delay Circuit Model,” Research Note, April 2026.

[3] Noriaki Kihara, “Realization of Simple Harmonic Oscillation and Sinusoidal Waves in the Delay Circuit Model,” Research Note, April 2026.